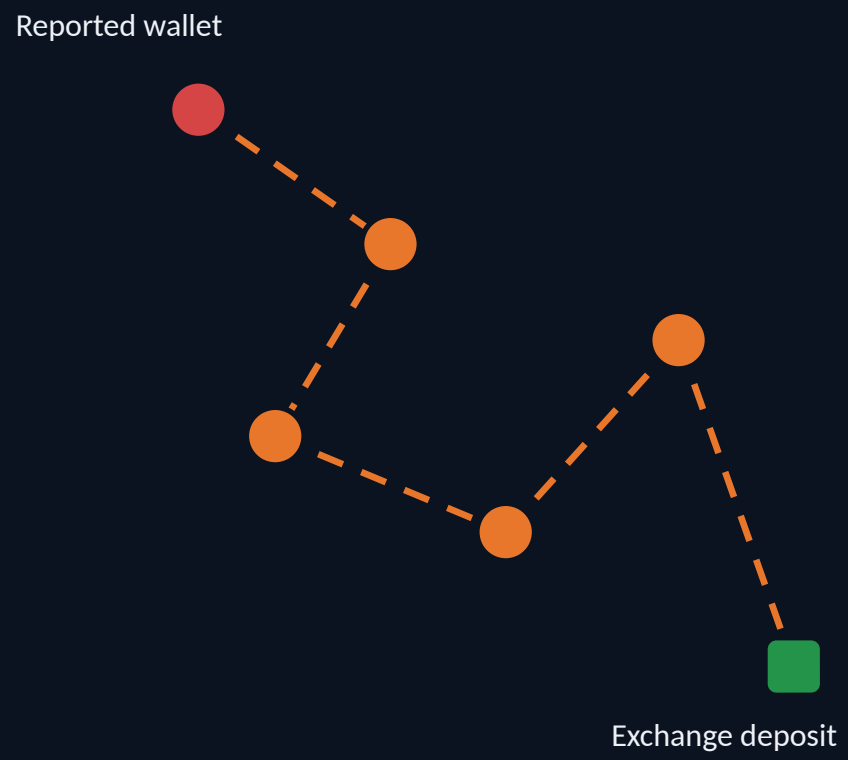
ChainSight

From a reported scam wallet to a ready-to-send freeze request, in one automated investigator report



Team [Team name] | Powered by TinyFish web agents

Team details

Team name: [Team name] Institution / organisation: [Name] Contact: [Email / phone]



[Member 1 name]

Team lead, product and pitch

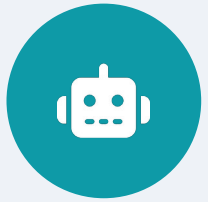
[College / year / LinkedIn]



[Member 2 name]

On-chain tracing and risk engine

[College / year / LinkedIn]



[Member 3 name]

TinyFish agents and backend

[College / year / LinkedIn]



[Member 4 name]

UI/UX and investigator report

[College / year / LinkedIn]

Roles are suggestions that match the work split; edit to fit your team.

Scam money moves in hours; tracing it takes days

Cyber-fraud victims report the wallet addresses scammers used. Those are usually burner or intermediary wallets that hide the trail (layering).

Investment scams | Task-based fraud | Sextortion | Phishing | Ransomware

How the money disappears



Meanwhile, the investigator is still researching

Why existing tools don't fully solve it: commercial tracing tools (Chainalysis, TRM and similar) are expensive and built for large firms, and even they don't automate the web research around a wallet or VASP contact discovery.

Which exchange received it?

Only exchanges (VASPs) can freeze funds and hand over identity. Finding them needs rare blockchain skills, across chains, mixers and bridges.

Who do we contact, and how?

Even once the exchange is known, finding its compliance contact and request process is manual web research.

Every hour counts

Funds move on within hours. Delay means the money is lost for good.

The gap: tracing tools stop at the web research

	Manual investigation	Commercial tracing tools	ChainSight
Cost	Low tooling cost, very labour-heavy	Expensive enterprise licences	**Low: public blockchain APIs + TinyFish**
Expertise needed	Rare blockchain experts	Trained analysts	**Guided report any investigator can use**
Finding the exchange	Hours to days	Fast	**Automated, down to the deposit address**
Web research on the wallet	Manual searching	Not automated	**Automated (TinyFish Search + Fetch)**
Exchange's law-enforcement contact	Manual web research	Not automated	**Automated (TinyFish Web Agent)**
Evidence for the case	Ad-hoc screenshots	Tool exports	**Timestamped, hashed page snapshots**

Comparison reflects the problem framing; ChainSight complements, not replaces, commercial tools for large firms.

Our solution: one wallet in, one investigator report out



A human investigator always reviews. The risk score is a lead, not proof of guilt. Every point in the score is explained, and every finding links to its evidence.

What the investigator gets: a case-ready report

Fund-flow graph

Every hop, amount and timestamp, with the path to the exchange highlighted

Nearest VASP + deposit address

Which exchange, how many hops, how much arrived, and when

Risk lead score + category

Rule-based, with the reason for every point (ML later)

Contact + draft freeze request

Portal, process, required documents, pre-filled request letter

Evidence trail

UTC timestamp and SHA-256 hash for each page relied on

PDF / JSON export

Standardised output for case files and other systems

Key insight

We find the deposit address, not just the exchange name

An unlabelled address that sweeps its funds into an exchange's hot wallet is that exchange's customer deposit address. That identifies the exact account the exchange can freeze and run KYC on.



scam wallet > hops > deposit > exchange

Built for the people who have to freeze the money

Primary

State cyber cells and police units

Teams working on national cybercrime complaints (NCRP) and SAHYOG

Secondary

Exchange and VASP compliance teams

Screen incoming deposits against reported fraud flows

Secondary

Banks' fraud units

Link crypto cash-outs to the fiat accounts they control

Value to them

Faster freezing

Exchange and contact identified in one run instead of days of manual work

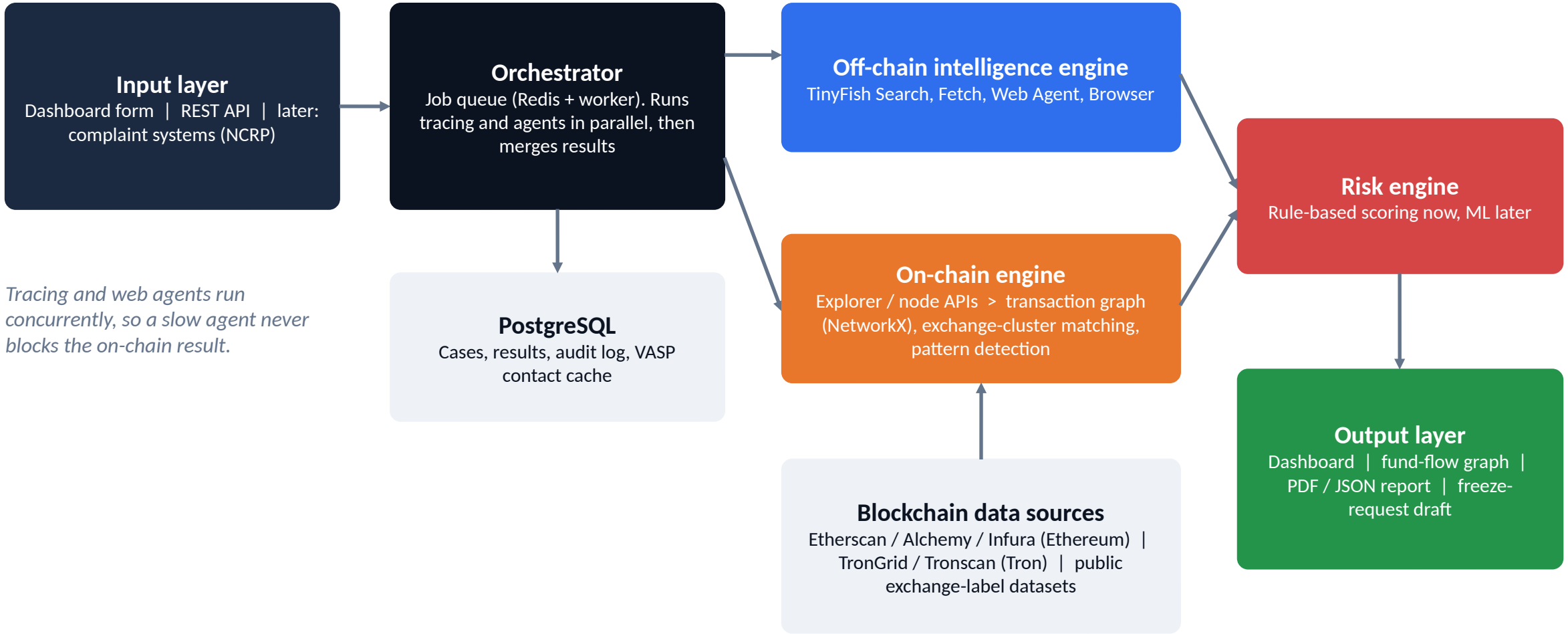
Less dependence on rare experts

Any trained investigator can act on the guided report

Standardised evidence

Same report format, hashed snapshots, audit log for every case

Technical architecture



Tracing and web agents run concurrently, so a slow agent never blocks the on-chain result.

Technology stack and MVP scope

Backend

Python + FastAPI

Async jobs

Redis + worker (Celery or similar)

Data

PostgreSQL for cases and results; NetworkX for the transaction graph

Frontend

React with a graph visualisation library for fund flow

Reports

PDF generation from case data, plus JSON

Web intelligence

TinyFish Search, Fetch, Web Agent, Browser (one API key)

MVP scope

- Ethereum and Tron only
- Single-wallet tracing
- Direct exchange identification
- Exchange contact discovery
- Everything else is roadmap

Why Tron: USDT on Tron is heavily used in Indian scam flows.

Blockchain data: Etherscan-style explorer APIs or Alchemy / Infura (Ethereum), TronGrid / Tronscan (Tron), Bitcoin explorers later. Free-tier limits and terms to be verified. Public exchange-label datasets are limited, and we say so.

How we integrate the TinyFish API

TINYFISH PIECE	WHAT IT DOES	WHERE CHAINSIGHT USES IT
Search	Structured live web results	Find mentions of the wallet on scam-report sites, forums and social pages
Fetch	Renders pages into clean text, JSON or HTML	Read those pages and confirm the address is really there; read exchange policy and contact pages
Web Agent	Natural-language goal in, multi-step task done, structured JSON out	Navigate exchange sites to the law-enforcement process and contact; pull P2P seller profiles
Browser	Cloud browser session we control	Timestamped evidence snapshots (stretch goal)

Why agents, not scrapers: these sites are all different, change often, have no APIs, and the tasks take several steps. One agent adapts to many exchanges without a script per site.

Honest limits: agents are slower than APIs and can fail on odd sites, so we use them only where no API exists, with retries and a manual fallback.

TinyFish in action: exchange contact discovery

1. We send the Web Agent a URL and a goal

"On this exchange's website, find the official process for law-enforcement agencies to request user information or freeze funds. Navigate legal, help-centre or compliance pages as needed. Do not log in or submit any form. Return the portal URL, process steps, contact email, required documents, response time and the pages you used."

2. It navigates the site and returns structured fields

Portal	Law-enforcement request page / form
Process	Official email domain > submit via portal > attach legal notice
Required documents	Signed notice on letterhead, officer ID, FIR / complaint ref.
Contact	Email if published, else portal only
Sources	Pages visited, each captured as hashed evidence

Retries with backoff

Contacts cached per exchange

Failures flagged for manual review

Simple HTTP API, one key (MCP also available)

Scalability and feasibility

Feasible now

Built on existing blockchain APIs and TinyFish. No custom blockchain indexer needed for the MVP.

Scales by configuration

A new chain or exchange is a config entry, not a new scraper. The agent finds each exchange's current page itself.

Parallel by design

Tracing jobs and agents run concurrently through the queue; contact results are reused across cases.

Risks and mitigations

Risk	Mitigation
Agent failures on unusual sites	Retries, cached results, fallback contact, flagged for manual review
False positives	Human validation; explained score; risk score is a lead, not proof
Legal and ethical limits	Authorised use only, audit logs, respect site terms; agents never log in or submit forms
Limited public exchange labels	Deposit-address inference from sweep behaviour; plug in richer label datasets

Use case: a task-scam victim's report, start to finish

1

Victim reports a wallet

A victim of a task-based job scam reports the USDT (Tron) wallet they paid.

2

Funds traced to an exchange

Money collected from several victims is split across mules and reaches an exchange deposit address within a few hops.

3

TinyFish agents work the web

They find outside reports of the wallet, the exchange's law-enforcement process and contact, and capture evidence.

4

Investigator acts

One report with a ready-to-send freeze and KYC request, reviewed and sent by the officer.

Result: hours or days of manual tracing and web research become one automated report. We will replace this with measured timings from live runs on real reported wallets.

Prototype: the investigator's case view

UI prototype shown on a sample scenario with synthetic wallet data.

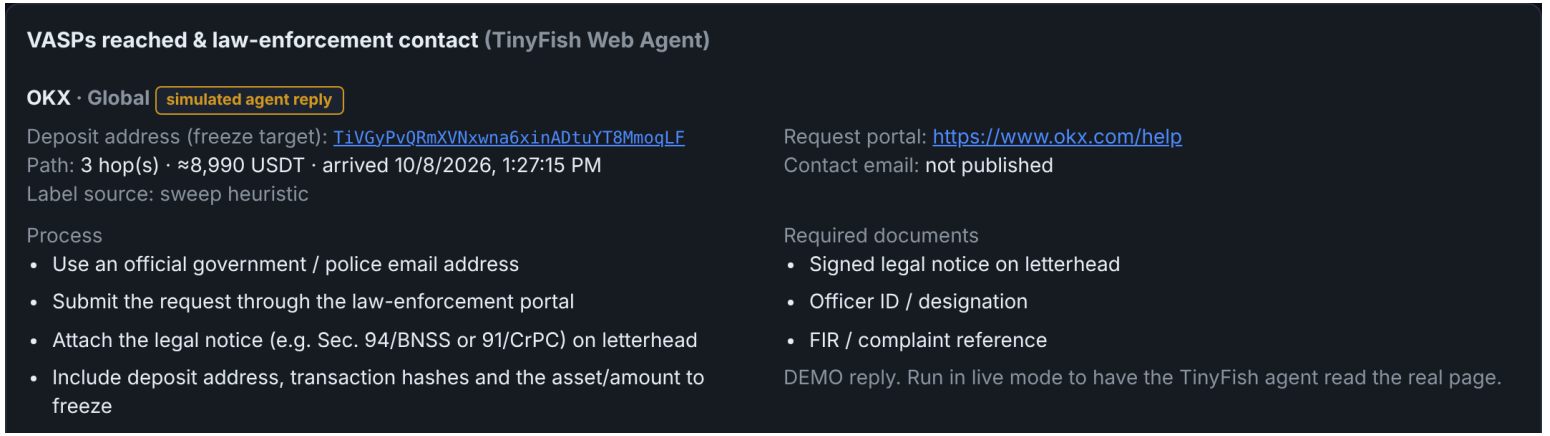


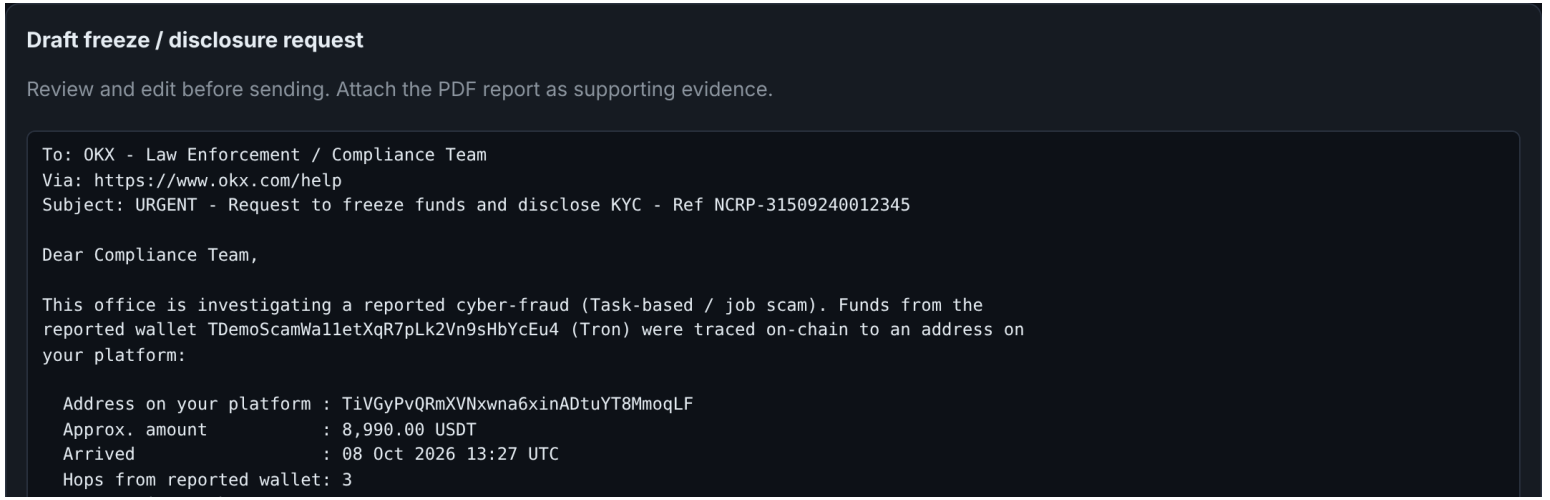
Reported wallet · Victim-side senders · Intermediaries · Exchange deposit address · Exchange wallet · Bridge

Highlighted path: funds from the reported wallet reach an exchange deposit address in 3 hops. Patterns detected: fan-in, pass-through, fan-out and a peel chain.

Prototype: exchange contact and freeze request

UI prototype on the same sample scenario; contact details are illustrative until the live agent runs.





Exchange contact card
Deposit address to freeze, path and amount, request portal, process steps and required documents found by the TinyFish agent.

Draft freeze and KYC request
Pre-filled with the complaint reference, deposit address, amount and transaction hashes. The officer reviews, edits and sends it.

Also in the prototype: confirmed wallet mentions, an evidence table with UTC timestamps and SHA-256 hashes, an itemised risk score and a PDF report.

Faster freezes, fewer losses

ChainSight turns a reported wallet into the exchange, the contact and the evidence an investigator needs to act while the money is still there.

Team [Team name] | [Contact email]

ROADMAP

- More chains (Bitcoin and others)
- Bridge and cross-chain tracing
- P2P seller intelligence module
- Address-poisoning guard
- ML-based risk scoring
- NCRP and SAHYOG integration